PyroRegen

by Team Smart Energy

"Turning Waste into Watts"

Project Proposal & Technical Brief

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1. Executive Summary

PyroRegen is a modular, locally-fabricated pyrolysis unit designed and developed by Team Smart Energy at the University of Ibadan. It is engineered to simultaneously address two of the most pressing crises facing Nigeria and Sub-Saharan Africa at large: the unmanaged accumulation of solid waste both plastic and agricultural residues and the persistent scarcity of affordable, clean energy across urban and rural communities.

At its core, PyroRegen works by thermochemically decomposing organic and plastic waste materials in a high-temperature, oxygen-free environment — a well-established industrial process called pyrolysis. Through this process, waste that would otherwise end up in open dumpsites or burn pits is converted into three commercially valuable outputs: bio-oil (a liquid fuel that can replace diesel and kerosene), biochar (a solid carbon-rich material that replaces charcoal and doubles as a soil enhancer), and syngas (non-condensable combustible gases that are recirculated back into the reactor to sustain its own temperature, dramatically reducing the external energy needed to run the unit).

The business case for PyroRegen is compelling. Each unit costs between ₦265,000 and ₦380,000 to produce and sells for ₦350,000 to ₦500,000, yielding a profit margin of 35 to 55 percent. More importantly, a single operational unit generates between ₦80,000 and ₦120,000 in monthly revenue from product sales alone, with an additional potential of up to ₦540,000 per month from carbon credit platforms such as Verra and Gold Standard — meaning a unit pays for itself within approximately 4 to 5 months of operation.

The total addressable market for waste-to-fuel solutions across Sub-Saharan Africa exceeds $10 billion, with the Nigerian bio-oil and biochar segment alone estimated at $60 million. Team Smart Energy's obtainable share, targeting 1,000 deployed units, is projected at $12 million.

Beyond the numbers, PyroRegen creates 3 to 5 local jobs per unit, reduces carbon dioxide equivalent emissions by 1.5 to 2 tons per ton of waste processed, and provides clean, affordable energy to communities that currently rely on smoky firewood, expensive kerosene, or unreliable diesel generators.

This document provides a comprehensive technical, operational, and commercial breakdown of the PyroRegen project — covering the science behind pyrolysis, the engineering of the unit, the business model, the financial projections, and the roadmap to scale. It is intended both as a reference for the team and as a submission-quality document for competition judges.

2. Problem Statement

2.1 The Dual Crisis Facing Nigeria and Sub-Saharan Africa

Nigeria and the broader Sub-Saharan African region are simultaneously grappling with two interconnected crises that are each significant on their own but devastating in combination: a waste management catastrophe and a chronic energy poverty problem. These two crises affect different segments of the population in different ways, but both ultimately trace back to the same failure of infrastructure and policy and both can be addressed with a single, well-designed solution.

The Urban Plastic Waste Crisis

According to the World Bank's What a Waste 2.0 report, Sub-Saharan Africa currently generates approximately 174 million tonnes of municipal solid waste every year. This figure is not static it is expected to triple to 516 million tonnes by the year 2050, driven by rapid urbanisation, population growth, and rising consumption of packaged goods. Nigeria sits at the epicentre of this problem. The country produces at least 32 million tonnes of solid waste annually, a number that is projected to climb to 107 million tonnes by mid-century.

The vast majority of this urban waste is plastic: sachets, packaging films, PET bottles, polythene bags, and single-use items of every description. Nigeria's waste collection infrastructure is severely underfunded and inconsistently managed, meaning a large proportion of this plastic never reaches a formal disposal site. Instead, it accumulates in drainage channels, floods streets during the rainy season, blocks rivers, and ends up in illegal open dumpsites where it is eventually burned. Open burning of plastic waste releases a cocktail of toxic chemicals into the atmosphere — including dioxins, furans, polycyclic aromatic hydrocarbons, and particulate matter — all of which are known to cause severe respiratory illness, cancer, and neurological damage in nearby communities.

The Rural Agricultural Residue Crisis

While urban Nigeria battles plastic mountains, rural Nigeria faces a different but equally serious waste burden. Nigerian agriculture particularly the cultivation of rice, maize, cassava, sorghum, and oil palm generates millions of tonnes of agricultural residues every year. These include rice husks, maize stalks, cassava peels, palm fronds, groundnut shells, and dried leaves. In most rural communities, there is no organised system for processing or utilising these residues. Farmers are left with no practical option other than piling them up and setting them alight in the fields — a practice known as open field burning.

Open field burning of agro-waste is wasteful on multiple levels. It destroys biomass that contains significant chemical energy, releases smoke and greenhouse gases that contribute to climate change, and degrades soil quality by burning off the organic matter that would otherwise enrich it. Meanwhile, the same rural communities that produce this waste suffer from acute energy poverty — they have little access to affordable cooking fuel, no reliable grid electricity, and are entirely dependent on expensive diesel generators for any mechanical power they need to run agro-machinery or irrigation pumps.

The Energy Poverty Dimension

Both urban and rural Nigerian communities share the burden of energy poverty. Access to clean, affordable cooking fuel is a persistent problem across the country. Millions of households still rely on firewood and charcoal for cooking — sources that are increasingly expensive as forest cover declines, and that produce indoor air pollution responsible for an estimated 93,000 deaths in Nigeria per year. Kerosene, once the default alternative, has become unaffordable for lower-income households following the removal of fuel subsidies. Grid electricity is unreliable even in urban centres, and rural communities are often entirely off-grid.

High fuel costs, unreliable access, and toxic indoor air pollution are the three faces of energy poverty in Nigeria — and they are directly linked to the waste crisis. The same waste that is clogging streets and burning in fields contains significant chemical energy that, if properly harnessed, could fuel the communities generating it.

2.2 Why Existing Solutions Fall Short

Diesel and kerosene are expensive, environmentally destructive, and subject to supply disruptions and price volatility. Grid electricity is inaccessible to large portions of the rural population and unreliable even where it exists. Charcoal causes deforestation and indoor air pollution. Compressed natural gas infrastructure does not reach most communities where it is needed most. None of these options addresses the waste problem — they only address the energy problem, and they do so poorly.

What Nigeria needs is not another fuel that must be imported or extracted — it needs a solution that transforms the waste already being generated into the energy that is already needed. That is precisely what PyroRegen delivers.

3. Our Solution — PyroRegen

3.1 What PyroRegen Is

PyroRegen is a modular, community-scale pyrolysis unit that converts plastic waste and agricultural residues into three commercially valuable products: bio-oil, biochar, and syngas. The unit is designed to be locally fabricated using materials available within Nigeria, operated by community members with basic training, and scaled from a single unit serving one community to hundreds of units networked across multiple states.

The name PyroRegen reflects two core ideas: pyrolysis (the thermochemical conversion process at the heart of the technology) and regeneration (the idea that waste is not an endpoint but a starting point — the raw material for regenerating energy, soil fertility, and economic activity in the communities that produce it).

3.2 The Three Outputs

Bio-Oil

Bio-oil is the primary liquid fuel output of the pyrolysis process. It is a dark brown, viscous liquid with a calorific value that makes it suitable as a direct replacement for diesel and kerosene in cooking stoves,

generators, and agricultural machinery. PyroRegen produces bio-oil from both plastic feedstock and agricultural waste, though the yield ratios and quality differ between the two — plastic pyrolysis typically produces a higher-quality, lighter bio-oil, while agro-waste pyrolysis produces a heavier oil with more organic acids. Both are usable fuels, and the product can be further refined with simple distillation if a higher-quality fuel is required. PyroRegen sells bio-oil at ₦250 to ₦350 per litre, compared to kerosene at ₦900 to ₦1,200 per litre — a cost saving of 60 to 70 percent for the end consumer.

Biochar

Biochar is the solid carbon-rich residue left behind in the reactor after pyrolysis. It is chemically similar to activated charcoal and has two distinct markets. As a cooking fuel, it can directly replace wood charcoal, burning with less smoke and higher heat output. As a soil amendment, it is even more valuable: biochar has a highly porous structure that improves soil water retention, reduces leaching of nutrients, and provides a long-term habitat for beneficial soil microbes. Applied to degraded agricultural soils, biochar can improve crop yields by 10 to 30 percent depending on soil type and application rate. PyroRegen sells biochar at ₦300 per kilogram, positioning it both to the household fuel market and to the growing market of smallholder farmers interested in soil improvement.

Syngas

Syngas or synthesis gas is the mixture of non-condensable combustible gases (primarily carbon monoxide, hydrogen, and light hydrocarbons) that remain after the heavier vapours have been condensed into bio-oil. Rather than venting or flaring this gas, PyroRegen's design routes it back through the reactor's heating system as a supplementary fuel. This syngas recirculation loop is a defining feature of the design: it means that after the initial startup heating phase, the reactor draws on its own byproduct to maintain operating temperature, reducing external energy input by an estimated 70 to 80 percent during steady-state continuous operation. This is discussed in full detail in Section 6.

3.3 Key Design Principles

PyroRegen was designed around four principles that reflect the specific context of its target market:

- Modularity: Each unit is self-contained and can be deployed independently. Multiple units can be operated in parallel at a single site to increase throughput. This means the system scales with demand rather than requiring a large upfront infrastructure investment.
- Local Fabrication: The unit's structural components — the reactor vessel, the shredder housing, the condenser coils, the collection tanks — are all fabricated from materials (steel, mild iron, standard fittings) available in Nigerian markets. This keeps production costs low and enables local repair and maintenance without dependency on imported spare parts.
- Community Operation: The unit is designed to be operated by people without advanced engineering training. A clear operating manual and a structured training programme ensure that community members can safely run, maintain, and troubleshoot the unit after a brief orientation.
- Dual Feedstock Capability: PyroRegen is designed to process both plastic waste and agricultural residues, either separately or in combination. This makes it useful in both urban and rural settings and gives operators flexibility to use whatever feedstock is most locally abundant.

4. How It Works — The Process Explained

The PyroRegen process follows a clearly defined sequence of stages, each of which serves a specific engineering function. Understanding this sequence is important both for operating the unit correctly and for explaining the technology credibly to potential customers, partners, and investors. The stages are: collection and sorting, shredding, drying, pyrolysis, condensation and separation, and product collection.

4.1 Stage 1 — Collection and Sorting

Waste feedstock is collected from community waste streams, partner organisations (supermarkets, waste management companies, farms), or directly from open dumpsites. Before processing, the feedstock is manually or mechanically sorted to separate plastic from agro-waste and to remove any materials that should not enter the reactor — particularly metals, glass, and high-chlorine plastics such as PVC. PVC is specifically excluded because its pyrolysis produces hydrochloric acid gas, which is corrosive to the reactor and harmful to operators. A colour-coded sorting protocol is included in the operator training to make this step straightforward even for non-technical personnel.

4.2 Stage 2 — Shredding

Sorted feedstock is fed into the shredder, which reduces particle size to a uniform range of approximately 5 to 20 millimetres. This step is critical for two reasons. First, smaller and more uniform particle sizes ensure that heat penetrates evenly throughout the feedstock during pyrolysis, preventing cold spots where material fails to react properly. Second, smaller particles have a higher surface-area-to-volume ratio, which increases the rate of pyrolysis reactions and improves overall conversion efficiency. The shredder is driven by an electric motor and is the unit's primary external energy consumer during operation — one of the reasons why reducing reactor energy consumption through syngas recirculation is so important to the overall energy balance of the system.

4.3 Stage 3 — Drying

Shredded feedstock passes through a drying stage before entering the reactor. This stage is essential because moisture in the feedstock is thermodynamically expensive: every gram of water that enters the reactor must be heated to evaporation temperature and then vaporised, consuming energy that would otherwise drive pyrolysis reactions. For agro-waste in particular — which can have moisture contents of 20 to 40 percent depending on season and storage conditions — moisture removal is critical to achieving acceptable oil yields. PyroRegen's drying stage targets a moisture content below 15 percent in the processed feedstock. The dryer uses waste heat recovered from the exhaust stream, ensuring that the energy cost of drying does not significantly add to the overall external energy requirement.

4.4 Stage 4 — Pyrolysis

The dried, shredded feedstock enters the pyrolysis reactor — the heart of the PyroRegen system. Pyrolysis is the thermochemical decomposition of organic material at elevated temperatures (400 to 550 degrees Celsius for PyroRegen's target operating range) in the complete absence of oxygen. The absence of oxygen is what distinguishes pyrolysis from combustion: rather than burning the material

and releasing all its energy as heat and CO2, pyrolysis breaks the long polymer chains of plastics and the cellulosic structures of biomass into shorter molecules, producing a mixture of vapours (which condense into bio-oil), solid carbon (biochar), and non-condensable gases (syngas).

The pyrolysis reactor in PyroRegen is a heated cylindrical vessel sealed against atmospheric oxygen. Temperature is maintained within the target range using a combination of the external heating element (during startup) and the syngas recirculation burner (during steady-state operation). A thermocouple monitors reactor temperature continuously, and a simple PID control system adjusts the heating input to maintain the setpoint. Residence time — the duration for which feedstock remains at pyrolysis temperature — is maintained at 30 to 60 minutes for slow pyrolysis, which maximises biochar yield. Fast pyrolysis (higher temperatures, shorter residence times) favours bio-oil yield but requires more precise temperature control and a continuous feed mechanism, features planned for the next generation of the unit.

Important technical note: The target temperature range of 400 to 550 degrees Celsius is specifically chosen to optimise the bio-oil yield while maintaining adequate biochar production. At temperatures below 400 degrees Celsius, pyrolysis reactions are incomplete. At temperatures above 600 degrees Celsius, secondary cracking reactions convert bio-oil vapours into additional syngas, reducing liquid yield. The 400 to 550 degree range therefore represents the practical sweet spot for a unit optimised for bio-oil production.

4.5 Stage 5 — Condensation and Separation

The hot vapour stream exiting the reactor passes through a condenser — a water-cooled heat exchanger that cools the vapours below their dew point, causing them to condense into liquid bio-oil. The condenser is one of the most important components in determining bio-oil quality and recovery efficiency: insufficient cooling allows lighter, more volatile fractions to escape with the non-condensable gas stream, reducing bio-oil yield. PyroRegen's condenser design uses a counterflow arrangement to maximise heat transfer efficiency.

After the condenser, the liquid bio-oil is collected in a sealed storage tank. The remaining gas stream — syngas — is separated and directed either to the syngas recirculation burner (to heat the reactor) or to a secondary afterburner that destroys any residual volatile organic compounds before exhausting. A simple gas-liquid separator at the condenser outlet prevents bio-oil carry-over into the gas stream.

4.6 Stage 6 — Product Collection

Bio-oil accumulates in the collection tank and is periodically drawn off for storage and distribution. Biochar is removed from the reactor vessel at the end of each batch cycle after the reactor has cooled sufficiently — or through a continuous discharge mechanism in more advanced configurations. The biochar is cooled, weighed, bagged, and either sold directly or stored for later distribution. Syngas that is not recirculated is combusted in the afterburner and exhausted safely.

5. Technical Specifications & Operating Parameters

The following table summarises the key technical parameters of the PyroRegen unit as currently designed. These figures are based on a combination of published pyrolysis literature, thermodynamic calculations, and the performance characteristics of the components selected for the design.

Parameter	Value / Description
Reactor Type	Batch-feed, externally heated, sealed cylindrical vessel
Operating Temperature	400 – 550 °C (optimised for bio-oil yield)
Atmosphere	Oxygen-free (inert) — nitrogen purge at startup
Residence Time	30 – 60 minutes (slow pyrolysis)
Feedstock Moisture Limit	< 15% by weight (enforced by pre-drying stage)
Feedstock Particle Size	5 – 20 mm (post-shredder output)
Daily Feedstock Capacity	~100 kg/day (40 kg plastic + 60 kg agro-waste)
Monthly Throughput	~2,500 kg/month (25 operating days)
Bio-Oil Yield (Plastic)	~40% by mass (approx. 4 L/day from 40 kg plastic)
Bio-Oil Yield (Agro-Waste)	~25% by mass (approx. 3 L/day from 60 kg agro-waste)
Biochar Yield	20 – 30% by mass (~5 kg/day from agro-waste fraction)
Syngas Yield	15 – 20% by mass (recirculated as reactor fuel)
Energy Self-Sufficiency	~70 – 80% during steady-state continuous operation
Primary External Energy	Electric shredder motor (startup heating uses LPG or syngas)
Dimensions (approx.)	1.8 m length × 0.9 m width × 1.5 m height (assembled)
Construction Materials	Mild steel (reactor), stainless steel (condenser), standard fittings
Fabrication Location	Locally fabricated in Nigeria

These specifications reflect the current design iteration. As simulation work (detailed in Section 15) progresses and prototype testing is conducted, specific values will be refined and updated. The figures above represent engineering targets grounded in published pyrolysis literature and thermodynamic first principles.

6. Syngas Recirculation — The Self-Sustaining Loop

6.1 What Is Syngas?

Syngas (synthesis gas) is the mixture of non-condensable combustible gases produced during pyrolysis. It consists primarily of carbon monoxide (CO), hydrogen (H_2), methane (CH_4), and small amounts of other light hydrocarbons. These gases do not condense at room temperature and therefore pass through the condenser without becoming bio-oil. In poorly designed pyrolysis systems, syngas is simply vented or flared — wasting a significant energy resource and creating a potential safety hazard.

6.2 The Recirculation Principle

PyroRegen's design routes the syngas stream from the condenser outlet back to a burner positioned at the reactor's external heating jacket. This burner combusts the syngas to produce the heat required to maintain the reactor at operating temperature. The principle is thermodynamically sound: the energy locked in the syngas was derived from the same feedstock that is being pyrolysed — it is a fraction of the feedstock's chemical energy being redirected to sustain the conversion process itself.

This is not perpetual motion, and it is important to understand the distinction. Perpetual motion would imply creating energy from nothing — a violation of thermodynamic law. What PyroRegen's syngas loop achieves is energy cascading: a fraction of the chemical energy released from the feedstock is captured as syngas and redirected to maintain the thermal conditions needed to release more energy from the remaining feedstock. The net energy balance is positive because the feedstock itself is the energy source — the system is simply using part of its own output to sustain its own operation, much like a wood fire that uses its own heat to dry the next piece of wood before it ignites.

6.3 Why 70 to 80 Percent and Not 100 Percent?

Online content — YouTube videos, social media posts — sometimes presents pyrolysis machines as fully self-powered. This claim requires careful scrutiny. In a true continuous-feed, optimised industrial pyrolysis plant operating at steady state on a high-energy feedstock like polyethylene, it is theoretically possible to approach near-full energy self-sufficiency for the thermal components of the system. However, claiming 100 percent self-sufficiency without qualification is misleading for several reasons.

First, the shredder motor requires external electrical power continuously. Second, startup heating — bringing the reactor from ambient temperature to 400 to 550 degrees Celsius before any syngas is being produced — always requires an external energy input, whether LPG, diesel, or grid electricity. Third, small-batch reactors like PyroRegen's current design operate in cycles rather than continuously, meaning there are cool-down and reheat phases between batches that reduce the average self-sufficiency over the full operating day.

A more accurate and defensible claim — and the one PyroRegen makes — is that syngas recirculation reduces external energy input to the thermal system by 70 to 80 percent during steady-state operation. This is a genuine and significant achievement that substantially reduces operating costs and makes the unit viable in locations without reliable fuel supply. Future iterations of the unit, particularly a continuous-feed design with thermal mass optimisation, could push closer to full thermal self-sufficiency

7. Safety & Emissions Management

7.1 Closed-System Reactor Design

The pyrolysis reactor is a fully sealed vessel. All vapour generated during the pyrolysis process is contained within the system and routed through the process pipeline — from the reactor to the condenser, from the condenser to either the bio-oil collection tank or the syngas line. At no point during normal operation is raw pyrolysis vapour released to the atmosphere. This is fundamental to both operator safety and environmental compliance. The sealing system uses high-temperature gaskets rated for the operating conditions, and all pipe connections are flanged and bolted rather than threaded to prevent vapour leakage under thermal cycling.

7.2 Afterburner and Thermal Oxidiser

Plastic pyrolysis — particularly of mixed or contaminated plastic waste — can produce trace quantities of dioxins, furans, and other persistent organic pollutants in the vapour stream. These compounds are formed when chlorinated plastics (such as PVC) are present in the feedstock, even in small amounts. While PyroRegen's sorting protocol is designed to exclude PVC from the feedstock, belt-and-suspenders engineering practice requires a secondary destruction mechanism.

PyroRegen incorporates an afterburner — a thermal oxidiser — downstream of the condenser. Any syngas that is not recirculated to the reactor heating system is passed through the afterburner and combusted at temperatures above 850 degrees Celsius with a residence time of at least two seconds. At these conditions, dioxins and furans are thermally destroyed to destruction efficiencies exceeding 99.99 percent. The afterburner exhaust — now consisting primarily of carbon dioxide, water vapour, and trace nitrogen oxides — is the only gas stream that exits the system to atmosphere.

7.3 Pressure Relief Valve

The pyrolysis reactor operates at modest positive pressure as vapours are generated, but the system is not designed as a pressure vessel in the engineering sense — it operates near atmospheric pressure with the condenser providing a continuous vapour draw. Nevertheless, blockages in the vapour outlet line, thermocouple failure, or operator error could theoretically allow pressure to build. A calibrated pressure relief valve (PRV) is installed on the reactor vessel as a non-negotiable safety component. The PRV is set to open at a pressure slightly above the maximum normal operating pressure and vents to the afterburner rather than directly to atmosphere, ensuring that even emergency relief gas is safely treated.

7.4 Condenser and Wet Scrubber

The water-cooled condenser is the primary mechanism for recovering bio-oil from the vapour stream. It is designed for high recovery efficiency — maximising the amount of condensable vapour that becomes saleable bio-oil rather than escaping as aerosol or gas. After the condenser, a secondary wet scrubber captures any residual particulates and water-soluble compounds from the gas stream before it reaches the afterburner. The scrubber uses a simple recirculating water bath and requires periodic water replacement — this is detailed in the operator maintenance schedule.

7.5 Operator Training and Safety Protocol

Every PyroRegen unit is delivered with a structured operator training programme covering five areas: pre-startup checks and safety inspection; startup and shutdown procedures; normal operating parameters and alarm thresholds; emergency procedures (including fire response and pressure excursion response); and routine maintenance tasks (including condenser cleaning, PRV testing, and biochar removal). Training is delivered on-site during installation and takes approximately one full day. A laminated quick-reference card summarising the key operating steps and emergency procedures is permanently mounted on the unit.

8. Yield & Performance Data

8.1 Yield Ratios by Feedstock Type

The yield ratios from pyrolysis — the proportions of bio-oil, biochar, and syngas produced per unit of feedstock depend primarily on the feedstock type, the operating temperature, and the residence time. The table below summarises the expected yield ratios for PyroRegen's two primary feedstocks at its target operating conditions.

Output	Plastic Feedstock	Agro-Waste Feedstock	Notes
Bio-Oil	40 – 60%	20 – 30%	Plastic yields higher, lighter oil
Biochar	5 – 15%	25 – 35%	Agro-waste yields more char
Syngas	25 – 40%	40 – 50%	Recirculated as reactor fuel
Loss/Water	~3 – 5%	~3 – 5%	Residual moisture & reaction water

8.2 Monthly Performance Projections — One Unit, 25 Operating Days

The following analysis projects the monthly throughput and revenue of a single PyroRegen unit operating 25 days per month at its design capacity. This is the foundational unit-economics calculation that underpins the entire business model.

Waste Type	Daily Input	Monthly Input	Bio-oil / Day	Monthly Bio-oil	Biochar/Mo	Revenue Est.
Plastic Waste	40 kg	1,000 kg	4 L	100 L	—	₦60,000
Agro-Waste	60 kg	1,500 kg	3 L + 5 kg char	75 L	125 kg	₦82,500
TOTAL / MONTH	**100 kg**	**2,500 kg**	**7 L**	**175 L**	**125 kg**	**₦142,500**

Bio-oil priced at ₦600/L average. Biochar at ₦300/kg. Revenue figures are gross product revenue before deducting operating costs. Profit margin after labour, maintenance, and input collection is estimated at 35 to 55 percent, giving a net monthly income of ₦80,000 to ₦120,000 per unit.

9. Competitive Analysis

PyroRegen does not operate in a vacuum — it competes for the same energy spend and waste management budget as existing solutions. The table below provides an honest, detailed comparison across the dimensions that matter most to PyroRegen's target customers.

Energy Source	Environmental Impact	Affordable	Scalability	Accessibility	Efficiency	Wa Imp
Diesel / Kerosene	High carbon emissions	No	Low	Medium	High	No
Charcoal	Deforestation risk	Yes	Medium	High	Low	No
Grid Electricity	Source dependent	No	High	Low (rural)	High	No
CNG	Lower emissions	Med	Low	Low	Medium	No
PyroRegen ★	Upcycles waste	Yes	High	**HIGH ✓**	Med-High	**Co it**

The most important differentiator in this table is the final column — Waste Impact. Every competitor in this space either ignores the waste problem or, in the case of charcoal, actively makes it worse through deforestation. PyroRegen is the only solution that simultaneously addresses the energy deficit and the waste crisis. This dual-impact positioning is the cornerstone of PyroRegen's competitive advantage and should be emphasised consistently in all external communications.

10. Target Market & Customer Segments

10.1 Market Sizing

The total addressable market (TAM) for waste-to-energy solutions across Sub-Saharan Africa exceeds $10 billion when accounting for the full waste-to-fuel potential of the region's growing organic and plastic waste streams. This figure reflects the combined market opportunity in bio-oil, biochar, carbon credits, and waste processing services across the continent.

Within Nigeria, the serviceable addressable market (SAM) for bio-oil and biochar products is estimated at $60 million, based on current household fuel expenditure patterns and the proportion of that spend that could realistically be redirected to bio-oil and biochar as substitutes for kerosene and charcoal respectively.

PyroRegen's initial serviceable obtainable market (SOM) — the realistic share capturable within a 3-year horizon with 1,000 operational units — is projected at $12 million. This represents approximately 20 percent of the Nigerian SAM, achievable through a combination of direct unit sales, waste collection contracts, and product sales.

10.2 Customer Segments — Who We Are Building For

PyroRegen serves five distinct customer segments, each with different motivations and procurement processes:

Waste Management Companies

Formal and informal waste management companies are sitting on a largely untapped revenue stream. They currently collect waste and pay to dispose of it at landfill sites or recycling aggregators. A PyroRegen unit allows them to convert their collected waste into saleable products — bio-oil and biochar — transforming a cost centre into a revenue source. This is a highly compelling value proposition that requires minimal behaviour change: they continue collecting waste, but instead of paying to dump it, they process it and sell the outputs.

Supermarkets and Retail Chains

Large retail operations generate significant volumes of plastic packaging waste daily. Many are under increasing pressure from consumers, regulators, and ESG (Environmental, Social, and Governance) compliance frameworks to demonstrate responsible waste management. A waste collection contract with PyroRegen allows them to document their plastic waste going to a productive end use rather than a

landfill, supporting their sustainability reporting. This segment is an attractive B2B partner for waste supply contracts.

Federal and State Governments

Federal and state agencies responsible for environmental management and energy policy are potential customers for both the units themselves (for deployment in community programmes) and for waste collection contracts at government facilities. The alignment between PyroRegen's outputs and Nigeria's national energy diversification agenda and its commitments under the Paris Agreement and the SDGs makes government procurement a realistic pathway, particularly if the team can secure an endorsement from a relevant regulatory or development body.

Smallholder and Commercial Farmers

Farmers represent a highly motivated customer segment for both bio-oil and biochar. Bio-oil provides affordable off-grid fuel for small irrigation pumps, threshers, and other agro-machinery — equipment that currently runs on expensive diesel. Biochar provides a low-cost soil amendment that can measurably improve yields on degraded soils. The appeal of a single product supplier who can take away the farm's agro-waste and return it as fuel and soil conditioner is strong, particularly in states like Oyo, Benue, and Nasarawa where both agro-waste volumes and fuel access challenges are significant.

NGOs and Climate-Oriented Programmes

International and domestic NGOs working on clean energy access, sustainable agriculture, or circular economy development represent a partnership and procurement channel rather than a direct commercial customer. These organisations can fund the deployment of PyroRegen units in target communities as part of their programme delivery, provide access to carbon credit markets, and offer introductions to government counterparts. Key alignment points include SDG 7 (Affordable and Clean Energy), SDG 12 (Responsible Consumption and Production), and SDG 13 (Climate Action).

11. Business Model & Revenue Streams

11.1 Core Unit Economics

The foundation of PyroRegen's business model is the unit itself. Each PyroRegen unit costs between ₦265,000 and ₦380,000 to fabricate, depending on component sourcing, batch size, and labour costs. Units are sold at a price of ₦350,000 to ₦500,000, yielding a gross profit margin of 35 to 55 percent on the equipment sale alone. This margin is achievable because all major structural components are locally fabricated, eliminating import duties and logistics costs that would otherwise inflate the price.

Customers who cannot afford the full purchase price upfront are offered an instalment plan: 20 percent of the purchase price is paid at the point of order, with the remaining 80 percent spread over 6 to 12 months. This lowers the barrier to adoption for smaller operators and community groups while maintaining a steady revenue stream for Team Smart Energy.

11.2 Revenue Streams

Stream 1 — Unit Sales and Installment Plans

Direct equipment sales are the primary revenue driver in the early phase of the business. Each sale generates immediate gross profit and creates a new operational unit in the field — which in turn generates ongoing demand for servicing, spare parts, and operator support, creating a secondary recurring revenue stream.

Stream 2 — Waste Collection Contracts

Once units are deployed, operators need a reliable supply of feedstock. PyroRegen can act as a waste broker, negotiating contracts with waste generators (supermarkets, factories, farms) who pay a monthly fee for their waste to be collected and processed. This is a recurring B2B revenue stream that provides predictable income independent of oil and biochar market prices. Monthly fees are projected at ₦15,000 to ₦30,000 per unit depending on waste volumes and logistics distances.

Stream 3 — Bio-Oil Sales

Bio-oil is sold at ₦250 to ₦350 per litre (average ₦300), compared to kerosene at ₦900 to ₦1,200 per litre. This price point — approximately one-third of kerosene — makes bio-oil immediately attractive to households and small businesses, while still generating a healthy margin for the operator. A single unit producing 175 litres of bio-oil per month generates approximately ₦52,500 in bio-oil revenue at the lower price point, rising to ₦61,250 at the higher end.

Stream 4 — Biochar Sales

Biochar is sold at ₦300 per kilogram. A single unit producing 125 kilograms of biochar per month from its agro-waste fraction generates ₦37,500 per month from this stream alone. The biochar market has two distinct channels: household cooking fuel (a direct replacement for wood charcoal) and agricultural soil amendment. The agricultural channel commands premium pricing — farmers who understand the soil improvement benefits of biochar are often willing to pay more than household fuel buyers.

Stream 5 — Carbon Credits

Carbon credits represent the highest-potential and most under-developed revenue stream in PyroRegen's model. By diverting waste from landfills and open burning — activities that would otherwise release significant quantities of methane and CO2 equivalents — PyroRegen generates avoided emissions that can be quantified and sold on voluntary carbon markets. The two most credible voluntary carbon standards are Verra (which operates the Voluntary Carbon Standard, or VCS) and Gold Standard. Both have methodologies for waste-to-energy and biochar projects.

A single PyroRegen unit processing 2,500 kilograms of waste per month can avoid an estimated 10 to 50 tonnes of CO2 equivalent per month depending on the baseline emission assumptions (landfill gas vs. open burning). At voluntary carbon market prices that currently range from $3 to $50 per tonne of CO2 equivalent, this translates to a potential of up to ₦540,000 per month per unit at favourable pricing — a number that, while optimistic at current small-scale pricing, represents a genuine and growing revenue

opportunity as carbon markets mature and PyroRegen builds a track record of verified emission reductions.

11.3 Payback Period

Based on the combined revenue streams above, a PyroRegen unit generating ₦100,000 to ₦142,500 in gross monthly product revenue, with operating costs of approximately ₦25,000 to ₦40,000 per month, achieves a net monthly income of ₦60,000 to ₦102,500. At a unit cost of ₦350,000 to ₦500,000, the payback period is approximately 4 to 5 months of operation. This is an exceptionally short payback period for a capital equipment purchase, and it is one of the most compelling numbers in PyroRegen's pitch to potential buyers and investors.

12. Financial Projections

12.1 Use of Seed Grant (₦500,000)

The initial ₦500,000 seed grant will be allocated to the following activities, in order of priority:

Activity	Allocation (%)	Amount (₦)
Prototype Development (reactor, condenser, shredder, controls)	40%	₦200,000
Component Procurement (valves, fittings, instrumentation)	25%	₦125,000
Marketing & Advertising (pitch deck, website, demo events)	15%	₦75,000
Field Testing (urban and rural pilot sites)	10%	₦50,000
Waste Collection Logistics (transport for pilot feedstock)	10%	₦50,000
TOTAL	**100%**	**₦500,000**

12.2 Three-Year Revenue Projection

The following projection models revenue growth across three phases of deployment, from a 5-unit pilot in Year 1 to 200 units in Year 3. Revenue includes unit sales, bio-oil sales, biochar sales, and waste collection contracts. Carbon credit revenue is excluded from this projection as it is subject to market verification timelines.

Metric	Year 1 (5 units)	Year 2 (50 units)	Year 3 (200 units)
Units Deployed	5	50	200
Revenue from Unit Sales	₦1.5M	₦15M	₦60M
Monthly Product Revenue	₦500k	₦5M	₦20M
Annual Product Revenue	₦6M	₦60M	₦240M
Estimated Total Annual Rev.	₦7.5M	₦75M	₦300M
Jobs Created	15 – 25	150 – 250	600 – 1,000
Waste Diverted / Month	~12.5 tons	~125 tons	~500 tons

These projections are estimates based on unit economics derived from published pyrolysis performance data and Nigerian market pricing. Actual results will depend on prototype performance, market adoption rates, and operational execution. The figures are intended to demonstrate the scale of opportunity, not to represent audited financial forecasts.

13. Impact Analysis

PyroRegen's impact operates across three dimensions that are interconnected and mutually reinforcing: environmental, social, and economic. Each dimension is quantifiable, and taken together they make a compelling case that PyroRegen is not simply a commercial opportunity but a genuinely transformative technology for the communities it serves.

13.1 Environmental Impact

Each operational PyroRegen unit diverts 1.2 to 1.5 tonnes of solid waste from landfills and open burn sites every month. Over the course of a year, a single unit removes 14 to 18 tonnes of waste from the environment — waste that would otherwise degrade slowly in a landfill, releasing methane as it does so, or be openly burned, releasing carbon dioxide, particulate matter, and toxic organic compounds.

The carbon accounting is significant. Pyrolysis prevents uncontrolled anaerobic decomposition of organic waste in landfills, which would otherwise generate methane — a greenhouse gas with a global warming potential approximately 28 times that of CO2 over a 100-year horizon. It also eliminates the open burning of plastic waste, which releases not only CO2 but persistent organic pollutants that have both local and global environmental impacts. The net emission reduction is estimated at 1.5 to 2 tonnes of CO2 equivalent per tonne of waste processed.

The biochar output provides a further environmental benefit: when incorporated into agricultural soil, biochar sequesters carbon for decades to centuries, providing a genuine carbon sink rather than merely avoiding emissions. Biochar also improves soil water retention and reduces the leaching of nitrogen and other nutrients, supporting more sustainable agricultural practices and reducing the need for synthetic fertilisers.

13.2 Social Impact

Each PyroRegen unit creates 3 to 5 direct local jobs covering waste collection, sorting, unit operation, product packaging, and local distribution. These are skilled-but-accessible jobs that can be filled by community members without advanced education after relatively brief training. At scale — 1,000 units — this translates to 3,000 to 5,000 direct jobs, plus a significant multiplier effect in the informal economy through waste collection networks and fuel distribution channels.

The energy access dimension of social impact is equally important. Households that currently spend ₦900 to ₦1,200 per litre on kerosene can access bio-oil at ₦250 to ₦350 per litre — a saving of 60 to 70 percent on cooking fuel costs. For a low-income household spending 10 to 15 percent of monthly income on cooking fuel, this saving is transformative. The reduction in indoor air pollution from switching from kerosene or firewood to bio-oil also has measurable health benefits, reducing the risk of respiratory illness, cardiovascular disease, and childhood stunting associated with household air pollution.

13.3 Economic Impact

Beyond the direct revenue and job creation at the unit level, PyroRegen creates new economic value chains in communities that currently have none. Waste collectors — many of them informal sector workers — gain a reliable, paying buyer for materials they collect. Local fabricators gain orders for reactor components. Farmers gain access to affordable fuel and a soil amendment that improves their yields. The bio-oil and biochar markets that PyroRegen creates are genuinely new markets in most of the communities it targets — not market share taken from competitors, but new economic activity generated where none existed before.

At the scale of 1,000 units processing 30 tonnes of waste per month each, PyroRegen would process 30,000 tonnes of waste monthly, produce more than 3 million litres of bio-oil, generate 37,500 tonnes of biochar, and avoid an estimated 45,000 to 60,000 tonnes of CO_2 equivalent every month. These are not marginal numbers — they represent a material contribution to Nigeria's waste management challenge and its energy transition goals.

14. Go-to-Market Strategy

PyroRegen's route to market is structured in four phases that reflect the progression from prototype validation to national scale. Each phase has specific objectives, activities, and success metrics, and each builds on the assets and learnings of the previous one.

Phase 1 — Awareness and Validation (Months 1 to 3)

The immediate priority is to build and test a working prototype. The prototype does not need to operate at full commercial capacity — its purpose is to demonstrate that the core process works as designed, to identify and resolve engineering issues before they become expensive problems at scale, and to generate real performance data (yield ratios, energy consumption, operating temperature stability) that replaces the estimates in this document with measured facts. During this phase, the team will also conduct community engagement in two pilot locations — one urban (Ibadan) and one rural (a farming community in Oyo State) — to validate the market assumptions around feedstock availability and product demand.

Phase 2 — Early Market Entry (Months 4 to 8)

With a validated prototype and real performance data, Phase 2 focuses on limited commercial deployment to key early adopter partners. These will be selected from the customer segments described in Section 10, prioritising those with the clearest and most immediate value proposition: waste management companies with existing collection infrastructure and farmers with both agro-waste supply and bio-oil demand. During this phase, waste collection contracts will be formalised, the first units will be delivered and commissioned, and the carbon credit registration process will be initiated with Verra or Gold Standard.

Phase 3 — Scaling and Distribution (Months 9 to 18)

Phase 3 targets the deployment of 50 units across at least three states, supported by small local fabrication hubs that can produce units closer to their deployment sites, reducing logistics costs and enabling faster delivery times. Local fabrication hubs will be established in partnership with existing metalwork and fabrication businesses, who will be trained to build PyroRegen units to specification. A franchise-style model will be explored for community operators, giving them exclusive territorial rights to the bio-oil and biochar markets in their area in exchange for a monthly royalty or product offtake agreement.

Phase 4 — Regional Expansion (Months 18 to 36)

By Month 18, with 50 or more units operational and a track record of performance data, PyroRegen will be positioned to seek Series A investment to fund regional expansion across West Africa. Ghana, Kenya, and Rwanda are identified as priority export markets based on their waste management challenges, their relatively business-friendly regulatory environments, and their active renewable energy and carbon market ecosystems. Phase 4 also includes an active pursuit of international climate finance through Green Climate Fund, African Development Bank, and bilateral development finance institutions.

Distribution Channels

The primary distribution channels are government and NGO procurement programmes (which can deploy units at scale to communities), local dealers and fabrication partners (who handle last-mile distribution and installation), direct B2B sales to waste management companies and large retailers, and participation in agrotech expos and trade fairs to build brand recognition and generate leads.

15. Simulation & Validation Plan

The simulation plan targets four tools, ranging from accessible to advanced, that together cover the full scope of what needs to be validated.

15.1 DWSIM — Process Flow Simulation (Start Here)

DWSIM is a free, open-source chemical process simulator with a graphical user interface that requires no programming experience to operate at a basic level. It includes thermodynamic property packages and unit operation models that are suitable for modelling the PyroRegen process from feedstock input to product output. The immediate goal with DWSIM is to build a steady-state process flow diagram that calculates the mass balance across the entire system: given a known feedstock composition and mass flow rate, how much bio-oil, biochar, and syngas does the process produce? This mass balance — backed by a DWSIM screenshot — transforms the yield figures in Section 8 from literature-derived estimates to simulation-validated projections. DWSIM is available free at dwsim.org.

15.2 Cantera — Reaction Kinetics

Cantera is a Python-based open-source toolkit for combustion chemistry and chemical kinetics. It allows the user to model the rates of pyrolysis reactions at different temperatures, which is the foundation for generating a temperature-versus-yield curve — a plot showing how bio-oil, biochar, and syngas yields change as the reactor temperature varies between 300 and 700 degrees Celsius. This curve is a powerful piece of technical evidence: it demonstrates understanding of the underlying chemistry and justifies the choice of 400 to 550 degrees Celsius as the operating range. Cantera is available at cantera.org and can be installed as a Python package.

15.3 Aspen Plus — Industry-Standard Process Simulation

Aspen Plus is the industry-standard process simulator used by professional chemical engineers globally. It has built-in models for biomass pyrolysis and gasification, and its thermodynamic property databases are among the most comprehensive available. An Aspen Plus simulation of the PyroRegen process would be the gold standard of validation at this stage of development. The software is commercially licensed, but AspenTech offers free academic licenses to university students and staff — the team should request a license through the University of Ibadan's engineering faculty. An Aspen Plus simulation, even a basic one, is the kind of technical detail that causes engineering judges to sit up and pay attention.

15.4 ANSYS Fluent — Reactor Geometry and Heat Transfer

ANSYS Fluent is a computational fluid dynamics (CFD) tool that models heat transfer, fluid flow, and chemical reactions in three-dimensional geometries. For PyroRegen, the relevant application is modelling how heat distributes across the reactor vessel and through the feedstock bed — identifying hot spots, cold spots, and optimal reactor geometry to maximise uniform heating. ANSYS offers a free student license at ansys.com/students that provides access to Fluent. A CFD simulation of the reactor, even a simplified 2D axisymmetric model, would represent a significant step forward in the technical sophistication of the project.

15.5 Simulation Milestone Goals

- A DWSIM process flow diagram showing the complete PyroRegen mass balance, with screenshot, to be included in the next version of this document and the competition presentation.
- A Cantera-generated temperature-versus-yield curve for plastic pyrolysis, showing bio-oil yield as a function of reactor temperature from 300 to 650 degrees Celsius.
- A basic Aspen Plus simulation confirming the energy balance of the syngas recirculation loop and quantifying the external energy savings achievable.

Recommended starting sequence: Install DWSIM first and build the process flow model. This will take approximately one to two weeks of focused work for someone with basic chemical process knowledge. The DWSIM output will also inform the Cantera model setup. Aspen Plus and ANSYS Fluent are higher-effort tools best tackled after the DWSIM work is complete.

16. Conclusion & Call to Action

PyroRegen represents a technically sound, commercially viable, and socially impactful solution to two of the most pressing challenges facing Nigeria and Sub-Saharan Africa. It is not a theoretical concept — the pyrolysis technology at its core is well-established at industrial scale globally, and the team has already developed a 3D engineering model of the unit, defined clear operating parameters, and built a credible commercial framework around it.

The work that remains before PyroRegen becomes an investor-ready project falls into three categories. First, prototype construction and testing: the team must build a working bench-scale or pilot-scale unit and generate real performance data. Second, process simulation: the yield figures and energy balance claims must be validated through DWSIM, Cantera, and Aspen Plus simulations, replacing literature estimates with team-generated data. Third, market validation: the team must engage directly with potential customers — waste management companies, farmers, community groups — to validate willingness to pay and refine the product offering based on real feedback.

None of these steps is beyond the reach of this team. The combination of agricultural engineering, civil engineering, and electrical and electronics engineering expertise on the team covers most of the technical disciplines needed to build the prototype and run the simulations. What is needed is focus, coordination, and a clear timeline — which this document is intended to support.

The vision is clear: PyroRegen turns the waste that is choking Nigeria’s streets and fields into the fuel that powers its homes and farms. One solution. Two crises. Real impact. This is worth building.

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